

SOC ANALYST L1

End-to-End Security Monitoring, Detection & Incident Response Lab

FORMAL PROJECT REPORT

Sysmon • Splunk • VirusTotal • Wireshark • PhishTool • Jira

A Practical Security Operations Center (SOC) Investigation Project

Prepared for: GitHub Portfolio / Entry-Level SOC Analyst Profile

Date: August 2026

DECLARATION

I hereby present this project report as a documentation of practical cybersecurity laboratory work focused on Security Operations Center (SOC) monitoring, detection, investigation and incident management. The work is intended for educational and professional portfolio purposes and was performed in a controlled environment.

ACKNOWLEDGEMENT

I acknowledge the official documentation and learning resources associated with the security tools used in this project. The project combines these tools into an integrated SOC workflow and presents the resulting procedures, queries, diagrams and investigation methodology as portfolio documentation.

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ABSTRACT

This project presents an end-to-end Security Operations Center (SOC) laboratory designed to demonstrate practical Level-1 security monitoring, detection, investigation, threat-intelligence enrichment, network analysis, phishing analysis and incident management. Six security tools are integrated: Sysmon for Windows endpoint telemetry, Splunk for SIEM-based analysis and alerting, VirusTotal for IOC enrichment, Wireshark for packet analysis, PhishTool for phishing investigation, and Jira for incident documentation and lifecycle management.

The workflow begins with security telemetry and proceeds through SIEM detection, L1 triage, evidence correlation, specialized analysis and incident documentation. The project covers failed logins, brute-force indicators, RDP activity, PowerShell execution, suspicious processes, network connections, DNS queries, file and registry activity, malicious hashes, suspicious IP addresses, domains and URLs, phishing emails, email authentication, packet analysis and structured incident records.

The key outcome is a reusable SOC investigation model showing that effective analysis depends on correlating multiple evidence sources rather than relying on a single alert. The report is designed both as formal project documentation and as a sanitized GitHub portfolio artifact.

Keywords: SOC, SIEM, Sysmon, Splunk, SPL, IOC, Threat Intelligence, Wireshark, Phishing, Incident Response, Jira

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1. INTRODUCTION

Organizations generate security telemetry from operating systems, endpoints, networks, applications, authentication systems and email. A SOC analyst must convert this high-volume telemetry into actionable findings. This requires monitoring, detection, triage, investigation, enrichment, documentation and escalation.

This project models those activities in a controlled lab. Sysmon supplies detailed endpoint events, Splunk provides centralized SIEM searches and alerts, VirusTotal enriches indicators, Wireshark provides packet-level evidence when captures are available, PhishTool supports email investigations, and Jira maintains the incident record.

1.1 Project Need

- Demonstrate practical SOC Analyst L1 capabilities.
- Connect endpoint, SIEM, network, email, threat-intelligence and ticketing functions.
- Practice evidence-based alert investigation.
- Create a structured and reproducible GitHub portfolio project.

1.2 Scope

- Windows endpoint monitoring using Sysmon.
- SIEM searching, reporting, dashboards and alerting using Splunk.
- IOC investigation using VirusTotal.
- Network and packet analysis using Wireshark.
- Phishing and malware-related email triage using PhishTool.
- Incident tracking using Jira.

2. PROBLEM STATEMENT AND OBJECTIVES

Tool-by-tool learning can leave a gap between recognizing an individual security event and performing a complete investigation. The project addresses this gap by defining a workflow in which alerts are investigated across several evidence sources and then documented as an incident.

2.1 Problem Statement

How can an entry-level SOC analyst detect suspicious activity, investigate its context, enrich relevant IOCs, correlate endpoint and network/email evidence, classify the finding and document the result in a repeatable way?

2.2 Objectives

- Collect and interpret endpoint telemetry.
- Write SPL searches for security monitoring.
- Build practical reports, dashboards and alerts.
- Investigate authentication, process, network, DNS, file and registry events.
- Investigate hashes, IPs, domains and URLs.
- Perform packet-level analysis when a capture exists.
- Analyze phishing emails and authentication indicators.
- Document findings and escalation in Jira.

2.3 Success Criteria

Capability	Expected Evidence
Detection	Repeatable SPL query and alert.
Triage	Host, user, timestamp, process and IOC context.
Enrichment	Threat-intelligence lookup of relevant indicators.
Network analysis	Relevant packet/protocol evidence where available.
Email analysis	Headers, SPF/DKIM/DMARC, links and attachments.
Documentation	Structured Jira incident with timeline and conclusion.

3. EXISTING SYSTEM

A basic security-monitoring environment may use separate tools for local Windows events, network captures, IOC reputation and ticketing. Although each tool is useful, the analyst may have to manually move between tools and construct the incident timeline.

3.1 Existing Characteristics

- Local event review is often reactive.
- Large event volumes make manual inspection inefficient.
- Endpoint and network evidence may remain disconnected.
- Threat intelligence may be consulted only after manual suspicion.
- Email investigations may not be linked to endpoint evidence.
- Incident records may not contain the complete investigation chain.

3.2 Limitations

Area	Limitation
Visibility	Security evidence is distributed.
Detection	Manual review does not provide consistent alerting.
Correlation	Events from different sources are difficult to connect manually.
Response	Evidence must be manually transferred into tickets.
Scalability	Repeated investigations become time-consuming.
Reporting	Monitoring information is not centralized.

3.3 Requirement

A more useful learning environment should preserve the strengths of individual tools while connecting them through a consistent SOC workflow.

4. PROPOSED SYSTEM

The proposed system integrates six tools around a central investigation process. Splunk serves as the SIEM and detection layer, while the other tools provide specialized evidence and enrichment.

4.1 Components

Component	Purpose
Sysmon	Endpoint process, network, file, registry and DNS telemetry.
Splunk	Central search, SPL, reports, dashboards and alerts.
VirusTotal	Hash, IP, domain and URL enrichment.
Wireshark	Packet capture and network protocol analysis.
PhishTool	Email header, authentication, URL and attachment analysis.
Jira	Incident documentation, workflow, escalation and closure.

4.2 Design Goals

- Repeatable investigations.
- Evidence traceability.
- Safe handling of sensitive data.
- Clear L1 analyst workflow.
- Ability to extend with additional data sources.

5. SYSTEM ARCHITECTURE

5.1 High-Level Architecture



Figure 1. High-level SOC architecture.

5.2 Evidence Correlation

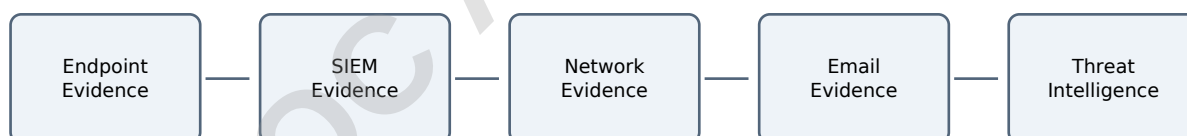


Figure 2. Evidence sources used during investigation.

The architecture is modular. Endpoint telemetry provides local context, Splunk centralizes detection, specialized tools add evidence, and Jira preserves the investigation record. The architecture is suitable for a lab and can later be expanded toward enterprise-scale monitoring.

6. PROJECT WORKFLOW



Figure 3. SOC investigation lifecycle.

6.1 Investigation Questions

- What happened?
- When did it happen?
- Which host and user were involved?
- Which process initiated the activity?
- What IP, domain or URL was involved?
- Is the behavior expected?
- What evidence supports the verdict?
- What action is required?

6.2 Evidence Timeline

Stage	Evidence
Detection	Alert, timestamp, rule.
Endpoint triage	Sysmon event, process, parent, command line.
Network triage	IP, port, protocol, DNS.
Enrichment	Hash/IP/domain/URL reputation.
Email triage	Sender, headers, SPF/DKIM/DMARC, URL, attachment.
Decision	Benign, suspicious, malicious or inconclusive.
Documentation	Jira timeline, evidence, action and closure.

7. TOOLS AND TECHNOLOGIES

7.1 Sysmon

Sysmon is the endpoint telemetry component used to record detailed Windows activity. Its events can support process, network, file, registry and DNS investigations.

7.2 Splunk

Splunk is the central SIEM layer. SPL searches filter and transform event data into investigation results, reports, dashboards and alerts.

7.3 VirusTotal

VirusTotal is used for IOC enrichment involving file hashes, IP addresses, domains and URLs.

7.4 Wireshark

Wireshark is used for packet capture analysis and display filtering. It supports protocol, field and value-based investigation.

7.5 PhishTool

PhishTool is used for phishing triage, email header analysis, authentication checks and suspicious URL/attachment review.

7.6 Jira

Jira is used as the incident-management layer for structured evidence, status, severity, actions, escalation and resolution.

7.7 Technology Relationship

Layer	Technology
Endpoint	Windows + Sysmon
SIEM	Splunk
Threat Intelligence	VirusTotal
Network	Wireshark
Email Security	PhishTool
Incident Management	Jira

8. SYSMON ENDPOINT MONITORING

Sysmon provides detailed telemetry for endpoint investigations. High-value fields include Event ID, image, command line, parent process, user, process identifiers, destination IP/port, target filename, registry target, DNS query and hashes, depending on event type and configuration.

8.1 Important Sysmon Events

ID	Event	SOC Use
1	Process Create	Execution, command line, parent process, hashes.
2	File Creation Time Changed	Possible timestamping.
3	Network Connection	Process-to-network correlation.
5	Process Terminated	Process lifecycle.
7	Image Loaded	DLL/module investigation.
8	CreateRemoteThread	Possible process injection.
10	ProcessAccess	Process access/injection context.
11	FileCreate	New files/payloads.
12	Registry Create/Delete	Registry changes.
13	Registry Value Set	Persistence/configuration.
14	Registry Rename	Registry manipulation.
15	FileCreateStreamHash	Alternate data streams.
16	Configuration Change	Monitoring tampering.
17	Pipe Created	Named pipe activity.
18	Pipe Connected	Named pipe activity.
19	WMI Filter	WMI persistence.
20	WMI Consumer	WMI persistence.
21	WMI ConsumerToFilter	WMI persistence.
22	DNS Query	Domain/C2 investigation.
23/26	File Delete	Deletion activity.
25/30	Process Tampering	Process manipulation.

8.2 Practical Filters

```
EventID=1
Image contains powershell.exe
CommandLine contains -enc or -encodedcommand
ParentImage is an Office application
EventID=3
EventID=11
EventID=13
EventID=22
DestinationPort=443
QueryName contains suspicious domain
```

8.3 Suspicious Process Patterns

Examples requiring contextual review include Office applications spawning PowerShell or command shells, script interpreters launched from unexpected parents, encoded PowerShell command lines, execution from unusual user-writable directories, and unexpected outbound network connections.

9. SPLUNK SIEM AND SPL

Splunk provides the central security-search and alerting layer. In this project, Sysmon and Windows security data are queried to identify authentication, process and network behavior.

9.1 Core SPL Commands

```
index=
host=
source=
sourcetype=
EventCode=
earliest=
latest=
search
fields
table
sort
rename
dedup
where
eval
regex
rex
stats
chart
timechart
top
```

9.2 Core Detection Queries

```
# Failed logins
index=* EventCode=4625
| stats count by user, host, src_ip
| sort - count

# Brute-force indicator
index=* EventCode=4625
| stats count by src_ip, user
| where count > 5

# RDP login
index=* EventCode=4624 Logon_Type=10
| table _time user host src_ip

# PowerShell
index=* EventCode=1 Image="*powershell.exe*"
| table _time host user Image CommandLine ParentImage

# Network
index=* EventCode=3
| table _time host user Image DestinationIp DestinationPort Protocol

# DNS
index=* EventCode=22
| stats count by QueryName
| sort - count
```

9.3 Reports and Dashboards

The project dashboard can contain panels for failed logins, successful logins, PowerShell executions, suspicious processes, network connections, DNS activity, top users, top hosts, event trends and active alerts.

10. DETECTION RULES, ALERTS AND INVESTIGATIONS

10.1 Failed Login Alert

```
index=* EventCode=4625
| stats count by src_ip, user
| where count > 5
| sort - count
```

A scheduled alert can be configured to identify repeated authentication failures. The analyst should validate the time pattern, source IP, targeted user, host and whether a successful login follows the failures.

10.2 Suspicious PowerShell Alert

```
index=* EventCode=1
| search Image="*powershell.exe*"
| search CommandLine="*-enc*" OR CommandLine="*EncodedCommand*" OR CommandLine="*DownloadString*"
| table _time host user Image CommandLine ParentImage
```

10.3 Alert Investigation Procedure

Step	Action
1	Record alert name, time and rule.
2	Identify host and user.
3	Review related Sysmon events.
4	Inspect parent/child process and command line.
5	Review network and DNS activity.
6	Extract and enrich IOCs.
7	Use Wireshark/PhishTool where relevant.
8	Assign severity and confidence.
9	Create or update Jira incident.
10	Document conclusion and next action.

11. VIRUSTOTAL IOC INVESTIGATION

Threat intelligence is used to add context to indicators discovered during an investigation. The main IOC types in this project are SHA-256/SHA-1/MD5 hashes, IP addresses, domains and URLs.

11.1 Hash Analysis

- Record the exact hash.
- Review detection/reputation information.
- Check file type and associated observations where available.
- Correlate the hash with the endpoint timeline.

11.2 IP Analysis

- Check reputation and malicious associations.
- Review related domains or files when available.
- Compare with Sysmon Event ID 3 observations.
- Do not treat reputation alone as proof of compromise.

11.3 Domain and URL Analysis

- Check reputation and detections.
- Review related infrastructure.
- Correlate with DNS or email evidence.
- Preserve the IOC in the incident record.

11.4 IOC Correlation



Figure 4. IOC enrichment and documentation workflow.

12. WIRESHARK NETWORK ANALYSIS

Wireshark provides packet-level evidence when a capture is available. It can be used to investigate source/destination addresses, protocols, ports, DNS requests, TCP behavior and conversations.

12.1 Display Filters

```
tcp
udp
dns
http
tls
ip.addr == 192.168.1.10
ip.src == 192.168.1.10
ip.dst == 8.8.8.8
tcp.port == 443
dns.qry.name
tcp.flags.syn == 1
tcp.flags.reset == 1
http.request
```

12.2 Investigation Matrix

Question	Evidence
Who communicated?	Source and destination IP.
Which protocol?	Protocol field.
Which service?	Destination port.
Which domain?	DNS query.
Was TCP established?	TCP handshake/flags.
What happened in session?	Packet sequence / stream.

12.3 Correlation with Sysmon

If Sysmon Event ID 3 identifies a destination IP and port, Wireshark can be used to search a corresponding packet capture by timestamp, source, destination and protocol. This creates network-level context for the endpoint observation.

13. PHISHTOOL — PHISHING AND MALWARE ANALYSIS

Phishing investigations start with preservation of the original message and extraction of evidence. The analyst examines sender and recipient information, headers, authentication results, URLs and attachments.

13.1 Email Authentication

Control	Purpose	Analyst Interpretation
SPF	Checks authorized sending infrastructure.	Failure is a signal requiring context.
DKIM	Verifies domain-level message signing.	Anomalies require further investigation.
DMARC	Evaluates authentication/alignment policy.	Failure can support spoofing suspicion.

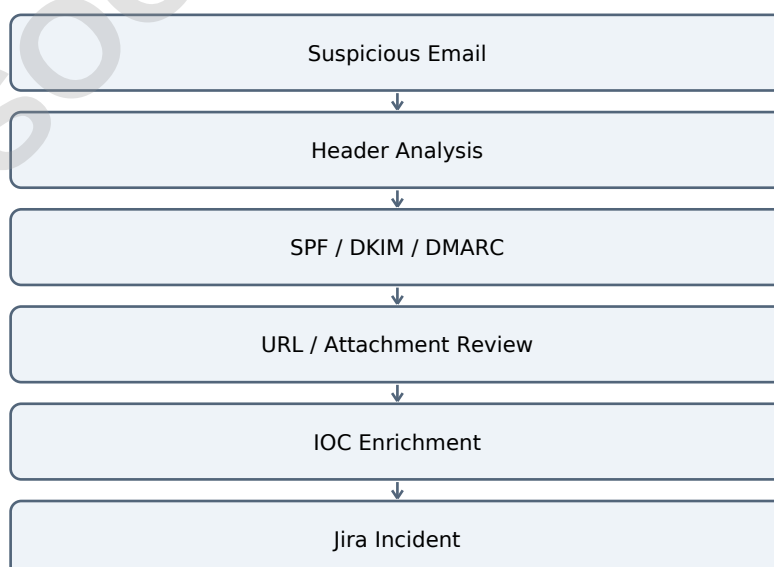
13.2 Phishing Indicators

- Look-alike sender domain.
- Urgent request for credentials or payment.
- Displayed-link/destination mismatch.
- Suspicious URL or redirect.
- Unexpected attachment.
- Authentication anomalies.
- IOC reputation indicating suspicious or malicious infrastructure.

13.3 Malware/Attachment Handling

Attachments should not be executed on a normal workstation. Record filename, type and hash when appropriate, enrich the hash using approved threat-intelligence resources, and preserve the findings in the incident record.

13.4 Email Investigation Workflow



14. JIRA INCIDENT MANAGEMENT

Jira is used as the case-management layer. The incident record should allow another analyst to understand what triggered the investigation, what evidence was collected, how the finding was classified, and what action was taken.

14.1 Incident Ticket Structure

Field	Example
Title	Multiple Failed Login Attempts Detected
Severity	Medium / High depending on evidence
Detection Source	Splunk
Affected Asset	Windows endpoint
User	Sanitized lab user
Timestamp	Event time
IOC	IP/domain/URL/hash
Evidence	Splunk, Sysmon, network/email evidence
Investigation	Timeline and analyst findings
Action	Monitor, contain, remediate or escalate
Status	Open → Investigation → Resolved → Closed

14.2 Incident Lifecycle



Figure 5. Incident lifecycle.

14.3 Documentation Rules

- Separate facts from assumptions.
- Use timestamps to build the timeline.
- Record exact IOCs where appropriate.
- Explain the reasoning behind the verdict.
- Document completed and recommended actions.
- Sanitize sensitive information before GitHub publication.

15. INTEGRATED INVESTIGATION SCENARIOS

15.1 Scenario A — Brute-Force Authentication

Splunk detects repeated Event ID 4625 failures. The analyst identifies the source IP, user and host, checks whether the activity is concentrated in a short period, and searches for successful authentication afterward. If suspicious, the source is enriched and a Jira incident is created.

15.2 Scenario B — Suspicious PowerShell

A PowerShell process is identified through Sysmon and Splunk. The analyst checks the command line, parent process, user and timestamp, then pivots to Event ID 3 and Event ID 22 to identify network and DNS behavior. Relevant IOCs are enriched and the findings are documented.

15.3 Scenario C — Phishing Email

A suspicious email is analyzed using PhishTool. Sender identity, SPF/DKIM/DMARC, URLs and attachments are reviewed. Extracted indicators are checked using threat intelligence and the verdict is recorded in Jira.

15.4 Scenario D — Suspicious Network Connection

A Sysmon network event shows an unexpected destination. Splunk provides process context, VirusTotal provides IOC enrichment, and Wireshark can validate packet-level evidence if a suitable capture exists.

15.5 Scenario E — Suspicious File Creation

```
index=* EventCode=11  
| table _time host user Image TargetFilename
```

The analyst reviews the file path, creating process and timestamp, obtains a hash where possible, and enriches it through threat intelligence.

16. IMPLEMENTATION AND EVIDENCE

The lab implementation is organized as controlled activities rather than production deployment. Evidence for the portfolio should include sanitized screenshots, query files, diagrams and incident examples.

16.1 Evidence Checklist

ID	Evidence
E01	Sysmon Operational event.
E02	Sysmon Event ID 1 process/command line.
E03	Sysmon Event ID 3 network connection.
E04	Splunk raw event/search result.
E05	Failed-login SPL and results.
E06	PowerShell detection and results.
E07	SOC dashboard.
E08	Alert configuration/result.
E09	VirusTotal hash analysis.
E10	VirusTotal IP/domain/URL analysis.
E11	Wireshark capture/filter.
E12	PhishTool email analysis.
E13	Jira incident ticket.
E14	Final timeline/closure.

16.2 Recommended GitHub Structure

```
SOC-L1-Security-Monitoring-Incident-Response/  
■■■ README.md  
■■■ 01-Sysmon/  
■■■ 02-Splunk/  
■■■ 03-VirusTotal/  
■■■ 04-Wireshark/  
■■■ 05-PhishTool/  
■■■ 06-Jira/  
■■■ diagrams/  
■■■ screenshots/  
■■■ Incident-Reports/
```

17. RESULTS AND DISCUSSION

The project demonstrates an integrated L1 investigation cycle. Endpoint events can be transformed into SIEM detections, alerts can be triaged, IOCs can be enriched, network/email evidence can be correlated, and the final decision can be preserved in an incident-management record.

17.1 Functional Results

Capability	Observed Project Outcome
Endpoint Monitoring	Process, network, file, registry and DNS activity can be investigated.
SIEM	SPL searches support filtering, aggregation and visualization.
Detection	Threshold and pattern-based alerts identify candidate incidents.
Threat Intelligence	Hashes, IPs, domains and URLs can be enriched.
Network Analysis	Wireshark filters isolate relevant packet activity.
Email Security	Phishing indicators and authentication results can be reviewed.
Incident Management	Jira provides a structured investigation record.

17.2 Analyst Skills Demonstrated

- Security-event interpretation.
- SPL construction.
- Alert triage.
- Endpoint process analysis.
- IOC extraction and enrichment.
- Packet analysis.
- Phishing analysis.
- Evidence-based severity assessment.
- Incident documentation and escalation.

17.3 Key Observation

Correlation produces stronger conclusions than isolated events. A suspicious PowerShell process becomes more significant when it has an unusual parent, creates a file, performs a suspicious DNS lookup and communicates with a suspicious destination. The analyst should still validate context before declaring compromise.

18. SECURITY CONSIDERATIONS

18.1 Data Protection

- Never publish passwords, API keys or tokens.
- Redact personal email addresses and usernames.
- Use lab accounts instead of production identities.
- Do not upload confidential enterprise logs.
- Treat suspicious files and URLs as potentially dangerous.

18.2 Evidence Integrity

- Preserve timestamps and source information.
- Separate raw evidence from interpretation.
- Record important queries exactly.
- Record IOC source and observation time.
- Use consistent incident identifiers.

18.3 Safe Analysis

Malware and suspicious attachments should be handled only in an authorized, isolated environment. Threat-intelligence services should be used according to their applicable terms and privacy requirements.

18.4 False Positives

Legitimate administration, automation, updates and security software can generate events resembling suspicious activity. L1 analysts should validate context before escalation.

19. LIMITATIONS

- Telemetry depends on Sysmon configuration and Windows environment.
- Splunk detections depend on correctly parsed fields.
- VirusTotal is enrichment, not standalone proof of compromise.
- Wireshark requires an appropriate packet capture.
- Email analysis depends on original headers/content.
- Jira manages incidents but does not detect threats.
- The lab does not claim complete ATT&CK; or enterprise coverage.
- Lab scale does not represent production SOC volume.

19.1 Scope Boundary

The project focuses on L1 activities: monitoring, triage, evidence collection, basic investigation, IOC enrichment, documentation and escalation. Advanced reverse engineering, enterprise SOAR and full digital forensics are outside the core scope.

20. FUTURE ENHANCEMENTS

20.1 Additional Data Sources

- Active Directory telemetry.
- Firewall and VPN logs.
- DNS server logs.
- Web proxy logs.
- Cloud identity/audit logs.
- EDR telemetry.

20.2 Detection Engineering

- Map detections to MITRE ATT&CK;
- Add risk-based thresholds.
- Create multi-source correlation searches.
- Implement allowlists/suppression for known-good activity.
- Measure false-positive rates.

20.3 Automation

- Automated IOC extraction.
- Automated enrichment.
- Alert-to-ticket integration.
- SOAR playbooks.
- Automated evidence collection with analyst approval.

20.4 Advanced Analytics

- UEBA.
- Authentication and network anomaly detection.
- Threat-intelligence feeds.
- Central case management.
- Detection performance metrics.

21. CONCLUSION

This project presents a practical SOC Analyst L1 environment integrating Sysmon, Splunk, VirusTotal, Wireshark, PhishTool and Jira. Each tool contributes a distinct evidence type while the overall workflow demonstrates how a security analyst moves from telemetry to detection, triage, enrichment, correlation, documentation and escalation.

The project demonstrates practical competencies in Windows event analysis, SPL, alert investigation, endpoint monitoring, IOC analysis, network packet investigation, phishing triage and incident management. It also establishes a foundation for future work in threat hunting, detection engineering, incident response and security automation.

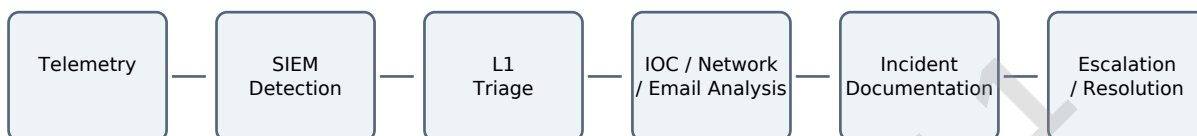


Figure 6. Final integrated SOC workflow.

22. REFERENCES

1. Microsoft Sysinternals — System Monitor (Sysmon) documentation, Microsoft Learn.
2. Microsoft Windows Security Auditing documentation, Microsoft Learn.
3. Splunk Documentation — Search Processing Language and Search Reference.
4. Wireshark User's Guide — Packet capture and display filtering documentation.
5. VirusTotal Documentation — Reports, IOC reputation/enrichment and API documentation.
6. Jira / Atlassian Documentation — Issues, workflows and project management.
7. PhishTool — Phishing analysis and email-security resources.
8. MITRE ATT&CK; — Enterprise techniques, tactics and detection knowledge base.

Reference URLs

Splunk: <https://help.splunk.com/>

Wireshark: <https://www.wireshark.org/docs/>

VirusTotal: <https://docs.virustotal.com/>

MITRE ATT&CK;: <https://attack.mitre.org/>

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APPENDIX A — SYSMON EVENT REFERENCE

ID	Event
1	Process Create
2	File Creation Time Changed
3	Network Connection
4	Sysmon Service State Changed
5	Process Terminated
6	Driver Loaded
7	Image Loaded
8	CreateRemoteThread
9	RawAccessRead
10	ProcessAccess
11	FileCreate
12	Registry Object Create/Delete
13	Registry Value Set
14	Registry Key/Value Rename
15	FileCreateStreamHash
16	Configuration Change
17	Pipe Created
18	Pipe Connected
19	WMI Event Filter
20	WMI Event Consumer
21	WMI ConsumerToFilter
22	DNS Query
23	File Delete
24	Clipboard Change
25	Process Tampering
26	File Delete Detected
27	File Block Executable
28	File Block Shredding
29	File Executable Detected
30	Process Tampering
31	Executable Detected
32	Imphash

Note: event availability and semantics depend on the Sysmon version and configuration used in the lab.

APPENDIX B — SPL QUICK REFERENCE

Use	Example
Basic search	index=*
Failed login	index=* EventCode=4625
Successful login	index=* EventCode=4624
RDP	index=* EventCode=4624 Logon_Type=10
PowerShell	index=* EventCode=1 Image="*powershell.exe"
Network	index=* EventCode=3
DNS	index=* EventCode=22
File creation	index=* EventCode=11
Registry	index=* EventCode=13
Statistics	stats count by host
Time chart	timechart count by EventCode
Filter	where count > 5
Fields	fields _time host user EventCode
Table	table _time host user EventCode
Sort	sort - count
Deduplicate	dedup user

Example Alert

```
index=* EventCode=4625
| stats count by src_ip, user, host
| where count > 5
| sort - count
```

APPENDIX C — WIRESHARK FILTERS AND JIRA TEMPLATE

C.1 Wireshark Filters

```
tcp
udp
dns
http
tls
ip.addr == 192.168.1.10
ip.src == 192.168.1.10
ip.dst == 8.8.8.8
tcp.port == 443
dns.qry.name
tcp.flags.syn == 1
tcp.flags.reset == 1
http.request
```

C.2 Jira Incident Template

```
Incident Title:
Date/Time:
Detection Source:
Severity:
Affected Host:
User:
Source IP:
Destination IP:
Domain / URL:
File Hash:
Detection Query:
Initial Observation:
Investigation Timeline:
Evidence Collected:
Threat Intelligence Findings:
Network Findings:
Email Findings:
Assessment / Verdict:
Actions Taken:
Escalation:
Recommended Follow-up:
Resolution:
Closure Date:
Analyst:
```

C.3 GitHub Publication Checklist

- Remove API keys, passwords and tokens.
- Redact personal/company-sensitive information.
- Use sanitized screenshots.
- Organize query files by tool.
- Include architecture and workflow diagrams.
- State that the environment is a controlled lab.
- Do not publish live malware or sensitive attachments.

APPENDIX D — PORTFOLIO SUMMARY

Project Title

SOC Analyst L1 — End-to-End Security Monitoring, Detection & Incident Response Lab

Core Toolchain

Layer	Tool	Evidence
Endpoint	Sysmon	Events, filters and endpoint investigations
SIEM	Splunk	SPL, alerts, reports and dashboards
Threat Intelligence	VirusTotal	Hash/IP/domain/URL enrichment
Network	Wireshark	Packet analysis
Email Security	PhishTool	Phishing and authentication analysis
Incident Management	Jira	Tickets, timelines and closure

Portfolio Statement

Designed and documented a practical SOC Analyst L1 environment integrating endpoint telemetry, SIEM detection, threat-intelligence enrichment, packet analysis, phishing investigation and incident management.



Figure 7. Recommended GitHub project structure.

END OF REPORT